Diabetes Prediction Model Report

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1. One-Line Summary

- Our team built an AI model that predicts "diabetes risk" using health checkup data.
- We confirmed practical performance using only the core features (`sex`, `age`, `HE_BMI`, `HE_wc`, `HE_glu`).
- Additional optional features (F1-F2) were conditionally applied to the app/server after separate simulation validation, and in production we suppress false positives (FP) by applying a blended decision threshold (`BLEND_THRESHOLD=0.54`).

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2. Problem We Solved

Question	Answer
What to predict?	Diabetes vs. non-diabetes (binary classification)
Data source?	KNHANES 2019 (`HN19_ALL.sav`)
Target age	Age 19 or older
Ground-truth label	Diabetes/non-diabetes split based on `HE_DM_HbA1c`

Original Data Source

Item	Details
Dataset name	Korea National Health and Nutrition Examination Survey (KNHANES), 8th cycle (2019-2021), 2019 raw data
Source file	`HN19_ALL.sav`
Source organization	Korea Disease Control and Prevention Agency (KDCA), KNHANES
Storage path in this project	`fastapi/resources/data/HN19_ALL.sav`
Reference document	`국민건강영양조사+제8기(2019-2021)+원시자료+이용지침서.pdf`

Why We Used This Data

Reason	Description
High-trust national data	National-scale health survey data with strong quality/reliability
Includes diabetes-related variables	Includes key variables such as age, BMI, waist circumference, and glucose
Suitable for Korean population prediction	Based on Korean samples, so it is better aligned with a domestic user app
Documented variable system	An official codebook exists, making variable interpretation and reproducibility easier

Ground-Truth (Target) Construction

Original Value	Mapped Value

Original Value	Mapped Value
1, 2	0 (non-diabetes)
3	1 (diabetes)

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3. Which Input Values (Features) Were Used?

3-1. Base Inputs

Column (Korean)	Description	Value Collected in App
`sex(성별)`	Sex	Sex
`age(나이)`	Age	Age
`HE_BMI(BMI)`	Obesity index (BMI)	BMI
`HE_wc(허리둘레)`	Abdominal obesity indicator	Waist circumference
`HE_glu(공복 혈당)`	Fasting glucose	Glucose (optional)

3-2. Additional Derived Features

Column (Korean)	How It Is Created	Why It Was Created
`HE_whr(허리-신장 비율)`	`HE_wc(허리둘레)` / `HE_ht(키)`	Better reflects body-shape characteristics
`HE_bmi_wc(BMI×허리 보정)`	`HE_BMI(BMI)` × (`HE_wc(허리둘레)` / 100)	Captures composite information beyond a single indicator

3-3. Two Experiments

Experiment	Number of Included Features	Characteristic
With glucose	7	Includes `HE_glu(공복 혈당)`
Without glucose	6	Predicts without `HE_glu(공복 혈당)`

3-4. Stepwise Data Size vs. Original (`HN19_ALL.sav`)

Step	Condition	Row Count (n)	Retention (vs. original)
Original	All records	8,110	100.0%
1st filter	`age(나이) >= 19`	6,606	81.5%
2nd filter	`HE_DM_HbA1c(당뇨 유병)` ∈ {1,2,3}	5,914	72.9%
3rd filter	Base feature validity (`HE_BMI(BMI)`, `HE_wc(허리둘레)`, `HE_ht(키)` > 0)	5,874	72.4%

3-5. Model Input DataFrame Info (Missing Status of Core/Extended Variables)

> Reference DataFrame: after 3rd filter, `n=5,874`

Variable	non-null	Missing Rate

Variable	non-null	Missing Rate
sex(성별)	5,874	0.00%
age(나이)	5,874	0.00%
HE_BMI(BMI)	5,874	0.00%
HE_wc(허리둘레)	5,874	0.00%
HE_ht(키)	5,874	0.00%
HE_glu(공복 혈당)	5,874	0.00%
F1_family_dm(가족력)	5,292	9.91%
F2_htn_or_med(고혈압/혈압약)	5,864	0.17%

> Note: HE_glu(공복 혈당) has some high-value outliers in the full dataset, and the app API uses an input range of 44-199.
> Among 5,874 base valid samples, 5,813 records fall within the app input range (44-199).

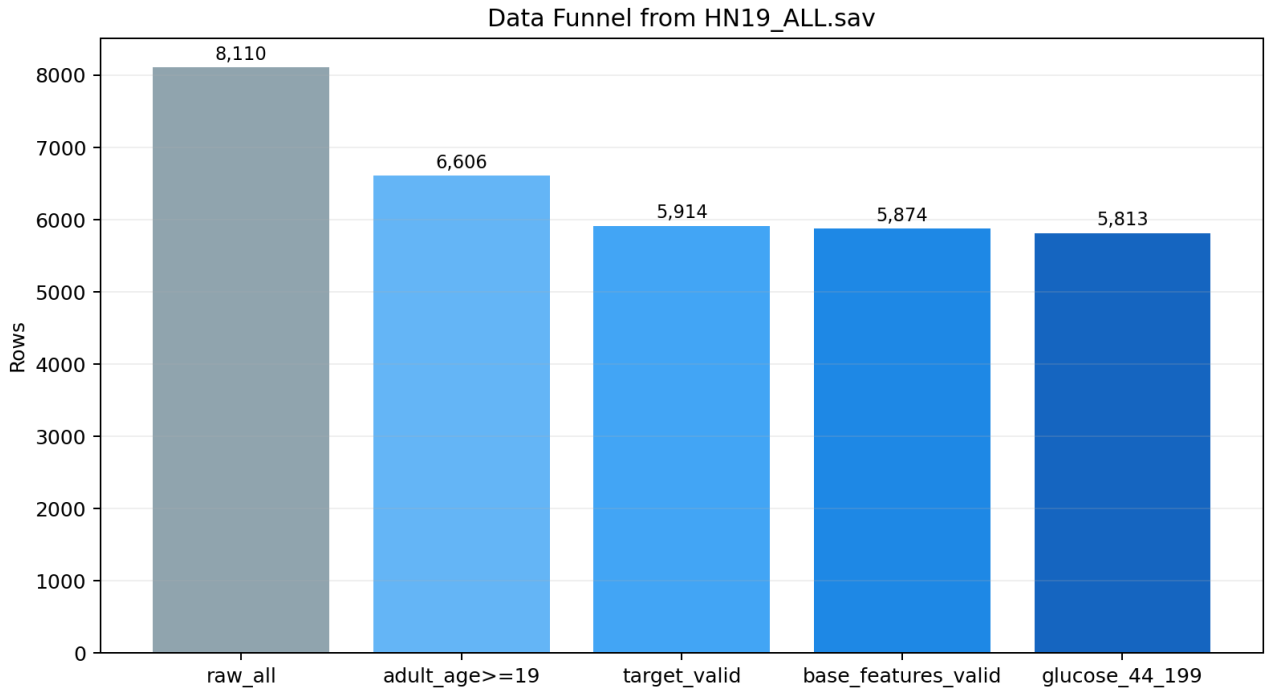
3-6. Summary Statistics of Core Continuous Variables (Describe, base valid sample n=5,874)

Variable	mean	std	min	25%	50% (median)	75%	max
age(나이)	51.57	16.78	19.0	38.0	52.0	65.0	80.0
HE_BMI(BMI)	23.92	3.59	13.98	21.47	23.60	25.99	50.29
HE_wc(허리둘레)	83.97	10.41	53.0	76.6	83.8	90.8	135.9
HE_ht(키)	163.69	9.36	133.3	156.7	163.3	170.5	194.0
HE_glu(공복 혈당)	101.23	22.42	53.0	90.0	96.0	105.0	339.0

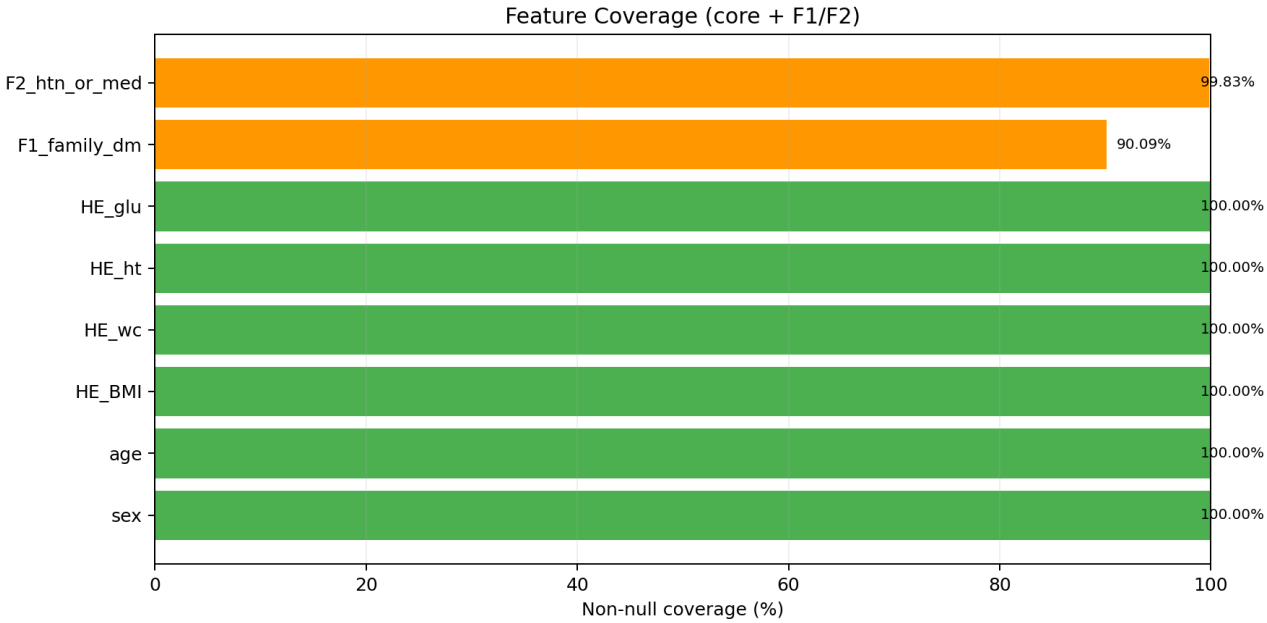
3-7. Core Features vs. Extended Candidate Features (From pre-validation/conditional-application perspective)

Category	Variable (Korean)	Initial Model (Core Features)	Extended Candidate Validation	Data Quality/Observation
Core (base)	sex(성별), age(나이), HE_BMI(BMI), HE_wc(허리둘레), HE_glu(공복 혈당)	Used	Kept as baseline features	Low missingness and stable
Extension F1	HE_DMfh1/2/3(가족 당뇨 진단력)	Conditionally used (no-glucose path)	Pre-validation + conditional application completed	~9.9% missing; auxiliary feature value confirmed in no_glu path
Extension F2	DI1_dg, HE_HPdr, HE_HP(고혈압/혈압약)	Conditionally used (priority application)	Pre-validation + conditional application completed	Very low missingness; improvements dominant in binned/exact paths

3-8. Data Status Visualization Charts


Data Funnel from HN19_ALL.sav
Rows
8,110
6,606
5,914
5,874
5,813
raw_all
adult_age>=19
target_valid
base_features_valid
glucose_44_199

-


Feature Coverage (core + F1/F2)
F2_htn_or_med
F1_family_dm
HE_glu
HE_ht
HE_wc
HE_BMI
age
sex
99.83%
90.09%
100.00%
100.00%
100.00%
100.00%
100.00%
100.00%
Non-null coverage (%)

-

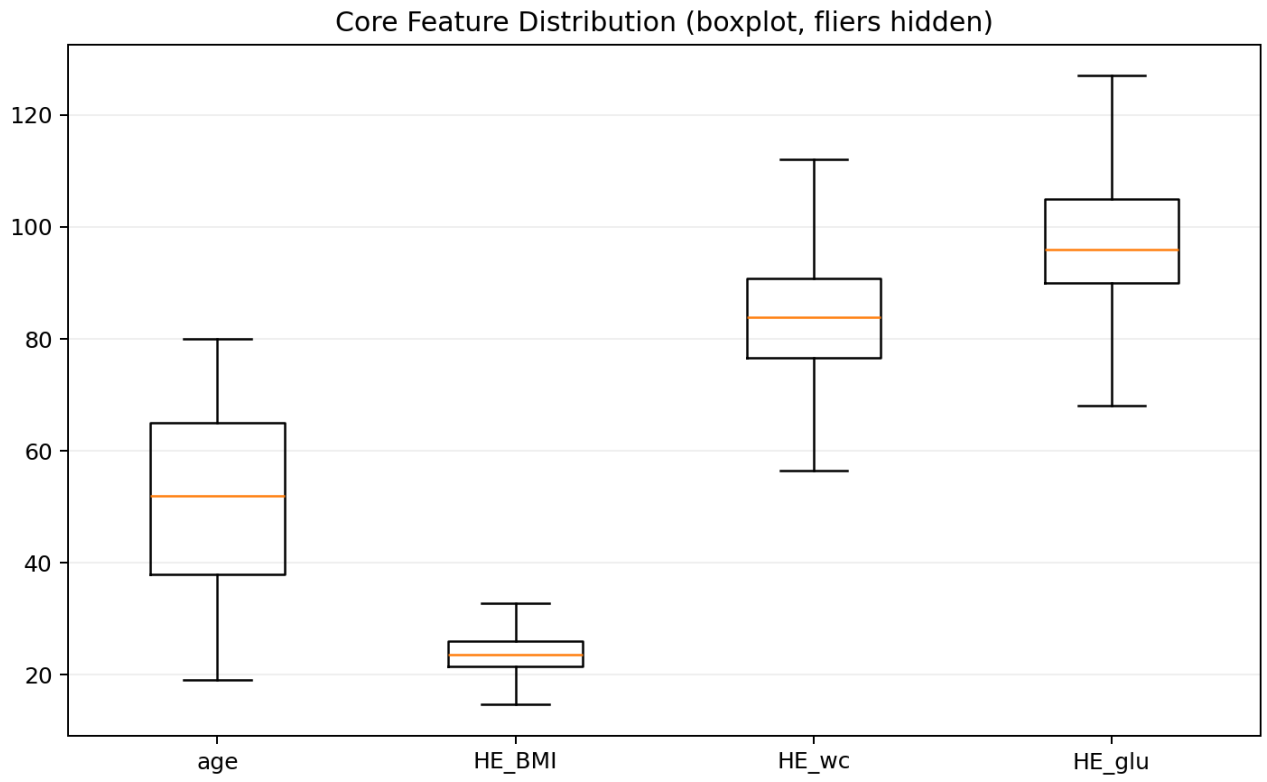


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Chart	Key Message
Data Funnel	Shows convergence from 8,110 original rows to 5,874 model-valid rows
Feature Coverage	Shows coverage distribution of core variables and F1/F2
Core Feature Boxplot	Provides intuitive ranges for `age(나이)`, `HE_BMI(BMI)`, `HE_wc(허리둘레)`, `HE_glu(공복 혈당)`

4. Data Preprocessing (Data Preparation) Pipeline

"Preprocessing" is the process of preparing data so the model can learn effectively.

Step	Action	Description
1	Keep adults only	Use age 19+ only
2	Label conversion	Simplify to diabetes/non-diabetes binary labels
3	Missing-value handling	Fill missing entries
4	Standardization	Bring values to a similar scale
5	Add derived features	Create new information such as waist-height ratio
6	SMOTE	Balance classes when diabetes samples are insufficient

Standardization vs. Normalization: Why Standardization?

Method	Meaning	Choice in This Project
Normalization	Scales values to 0-1	Not selected
Standardization	Scales to mean 0, std 1	**Selected**

Reason for selection: Distance-based models such as KNN were used, and standardization was more stable.

5. Train/Validation/Test Split

Data Split	Ratio	Role
Train	70%	Model training
Validation	10%	Threshold tuning
Test	20%	Final performance evaluation

In simple terms:

- **Train**: used for fitting the model
- **Validation**: used for threshold tuning
- **Test**: used for final performance evaluation

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6. Which Models Were Compared, and Why Was This Model Selected?

6-1. Compared Models

Category	Models
Linear	LogisticRegression, SGDClassifier
Tree/Ensemble	RandomForest, GradientBoosting, AdaBoost
Distance/Margin	KNN, SVM
Neural network	MLPClassifier

6-2. Final Selection

In both experiments, **KNN** achieved the best score.

Experiment	Final Model	Key Hyperparameters
With glucose	KNN	`n_neighbors=3`, `weights=distance`, `p=2`
Without glucose	KNN	`n_neighbors=3`, `weights=distance`, `p=2`

6-3. Hyperparameter Selection Process

- Compared parameter combinations using Grid Search.
- Scoring criterion: `balanced_recall`
  (reflects both overall accuracy and patient detection ability [Recall])

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7. Performance Results (Numbers)

7-1. Key Metric Comparison

Metric	With glucose (base+`HE_glu`)	Without glucose (base only)
Accuracy	0.8368	0.6675
Precision	0.4516	0.2267
Recall	0.8284	0.6036
F1	0.5846	0.3296

Metric	With glucose (base+`HE_glu`)	Without glucose (base only)
ROC-AUC	0.8608	0.6565

7-2. Why Accuracy Alone Is Not Enough

In medical problems, "not missing true patients" is critical.
So we placed special emphasis on **Recall**.

Metric	Meaning	Why It Matters
Accuracy	Overall correct prediction ratio	Reference only
Precision	Among predicted diabetes cases, true positive ratio	Warning precision
Recall	How well actual diabetes is found	Very important in medical settings
F1	Balance between Precision/Recall	Overall judgment

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8. Confusion Matrix (Prediction Summary)

Experiment	TN	FP	FN	TP
With glucose	880	170	29	140
Without glucose	731	348	67	102

Interpretation:

- Fewer **FN (missed patients)** is better for medical use.
- The glucose-included model is safer because it has fewer FN.

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9. Which Model Is Better for Medical Use?

Conclusion: The **KNN model with glucose** is more suitable.

Reasons:

1. Higher Recall (finds patients better)
2. Higher ROC-AUC (better overall discrimination)
3. Fewer FN in the confusion matrix (fewer missed patients)

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10. Do We Need All Columns in the App Implementation?

Conclusion: **No. Only core inputs are required.**

Category	Items (Variables)	Description
Direct app input	Sex (`sex`), age (`age`), height (`height_cm`), weight (for `HE_BMI` calculation), waist circumference (`HE_wc`)	Entered directly by user in the app
Optional app input	Glucose (`HE_glu`)	If provided, model with glucose is used
App-side calculation	BMI (`HE_BMI`)	Calculated from height/weight in app and sent to server

Category	Items (Variables)	Description
Server-side calculation	`HE_whr(허리-신장 비율)`	`HE_wc(허리둘레)` / `HE_ht(키)`
Server-side calculation	`HE_bmi_wc(BMI×허리 보정)`	`HE_BMI(BMI)` × (`HE_wc(허리둘레)`/100)

> Production input policy: `HE_wc(허리둘레)` remains a **required input** in this app.

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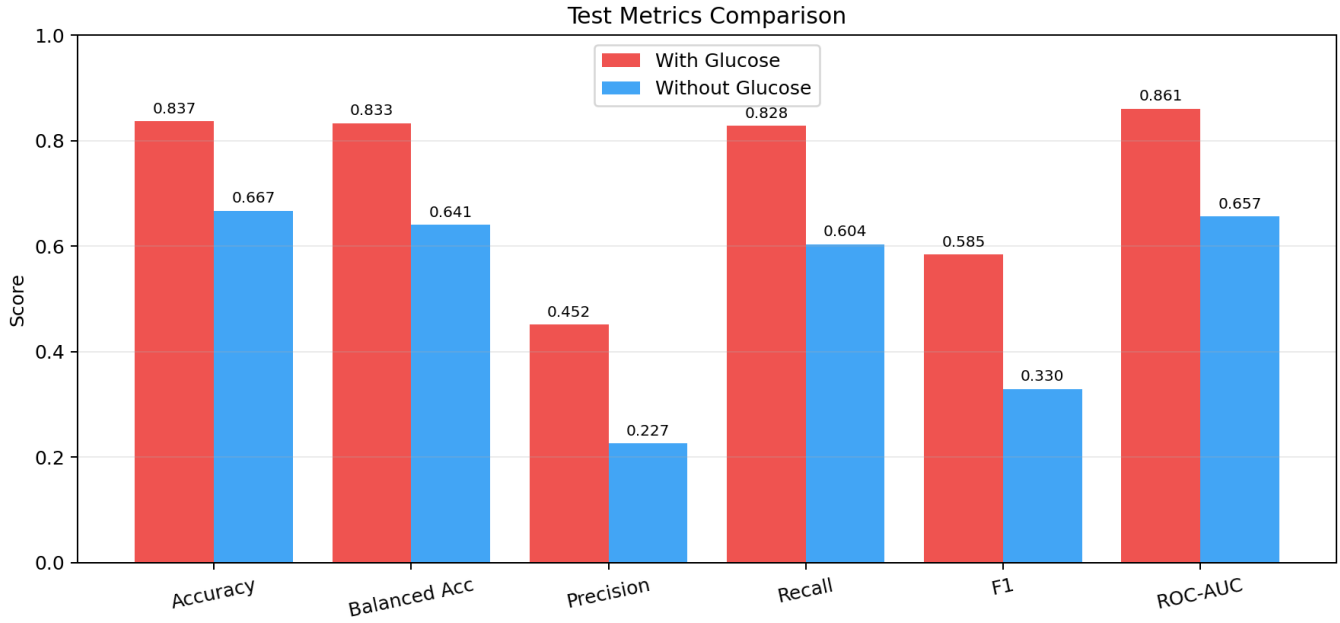
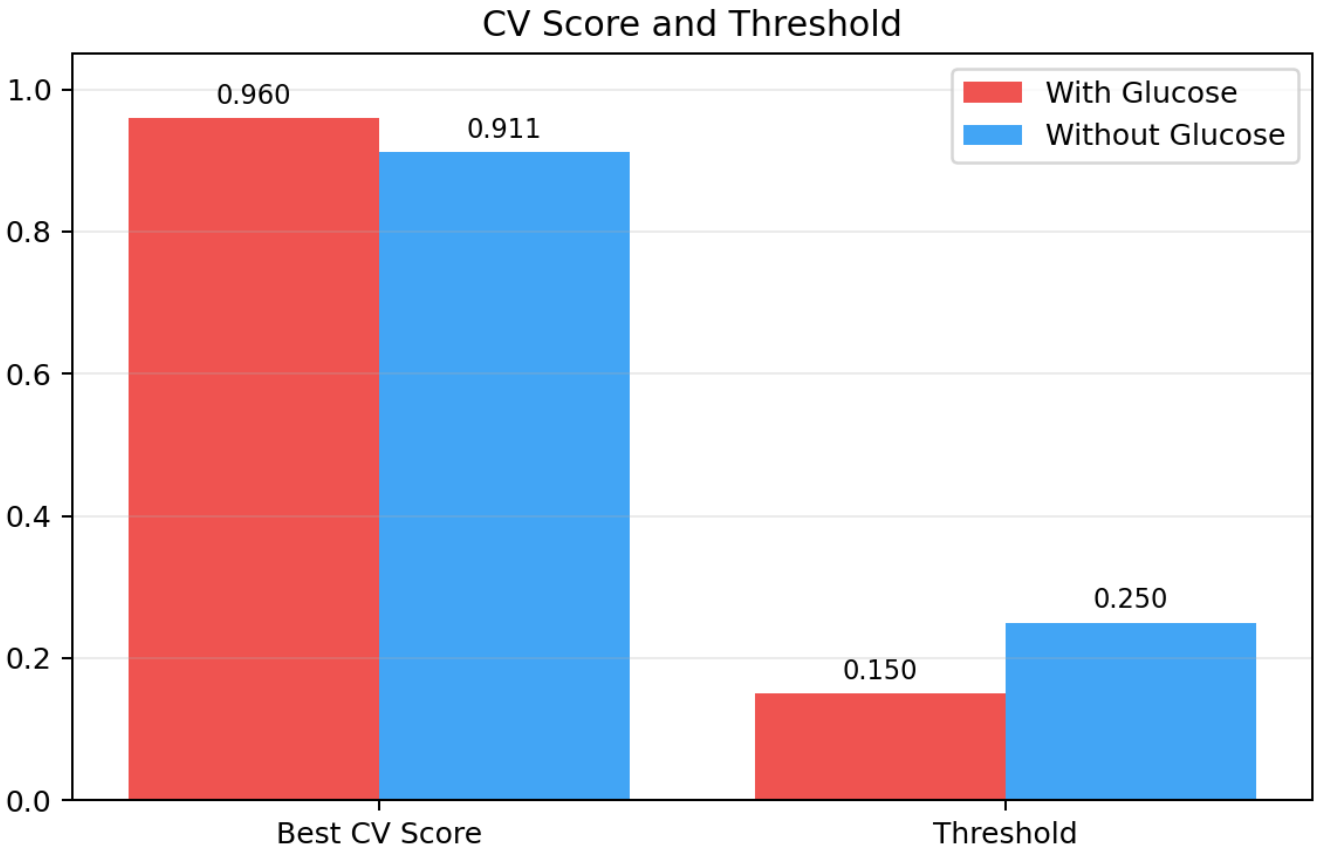
11. Results in Figures (Visualization Summary)

Figure 1. Test Metric Comparison



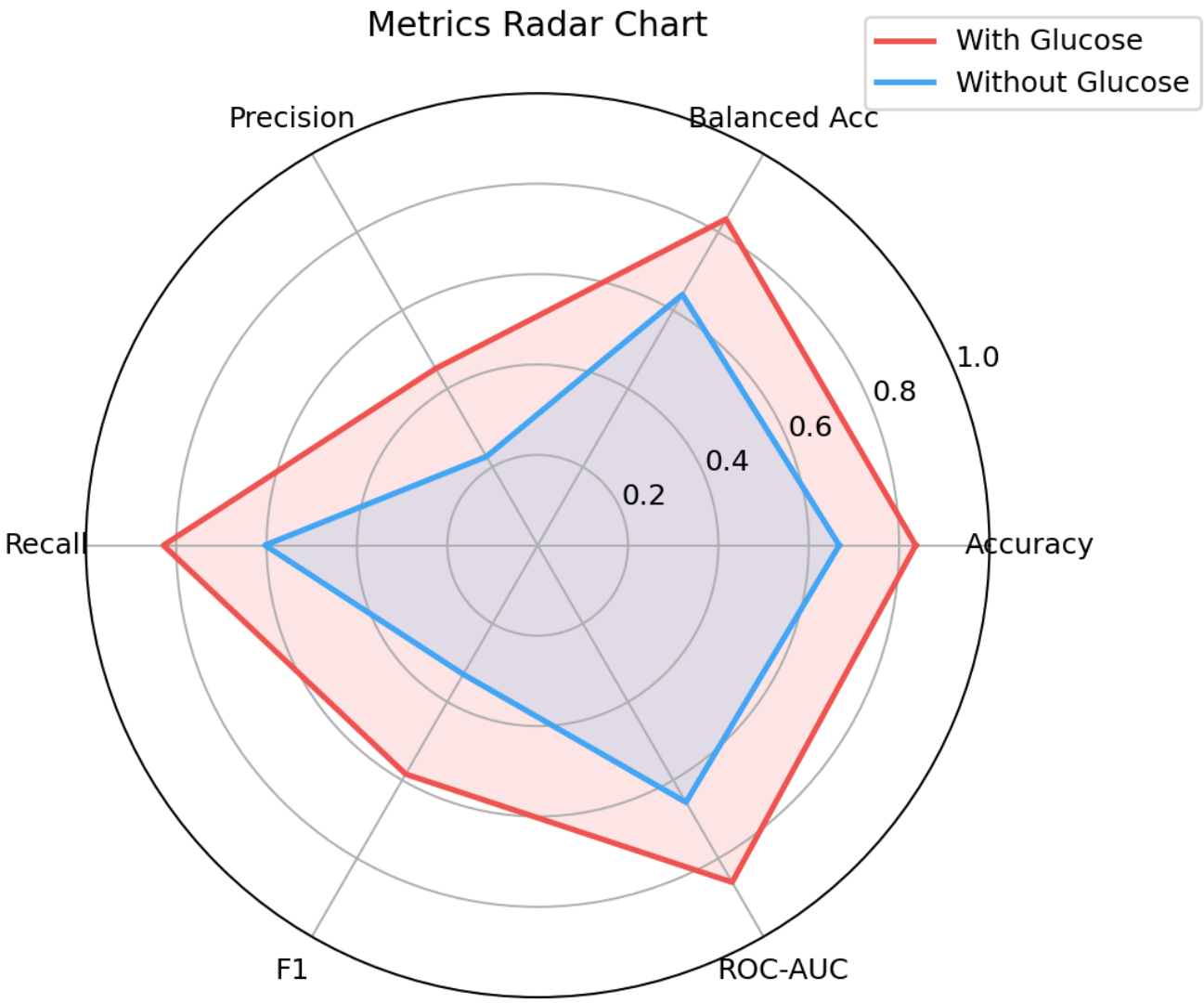
- This chart compares key performance metrics across two scenarios (with/without glucose).
- X-axis: metric type (Accuracy, Precision, Recall, F1, ROC-AUC)
- Y-axis: score (0-1)
- Interpretation point: **higher bars indicate better performance**. The with-glucose bars are higher for most metrics.

Figure 2. Cross-Validation (CV) Score and Threshold Comparison



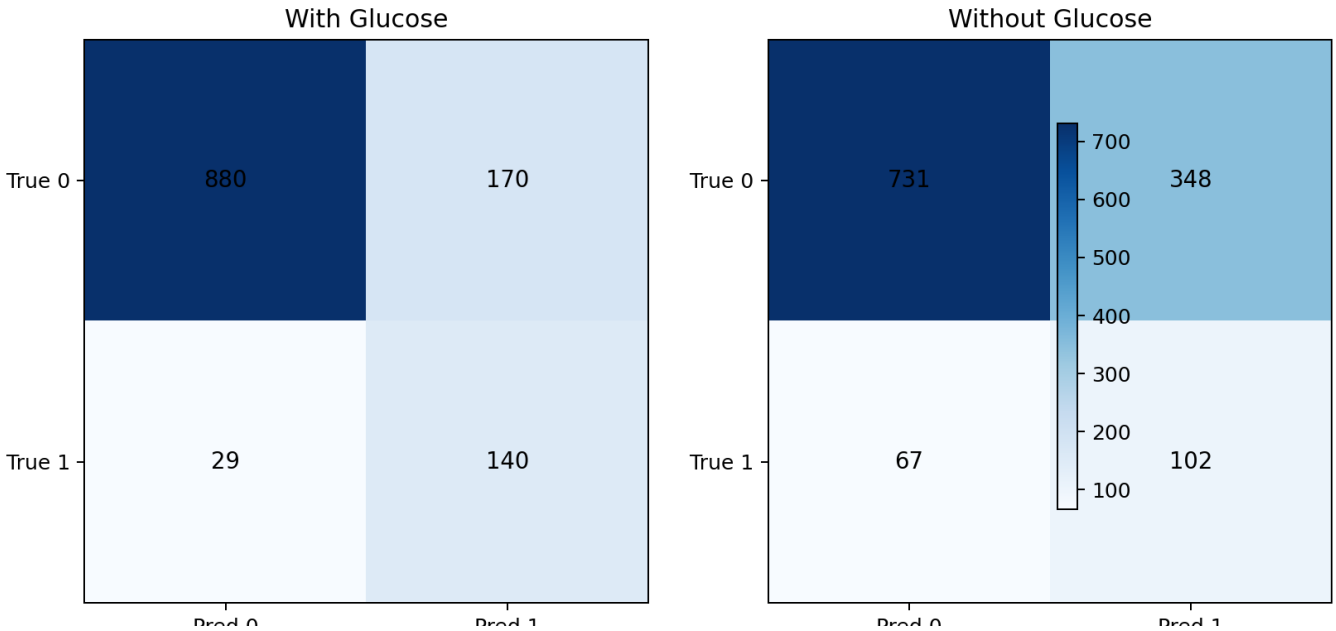
- This chart compares the CV score used for model selection and the final threshold.
- X-axis: `Best CV Score`, `Threshold`
- Y-axis: value (0-1)
- Interpretation point: higher CV score is better; threshold meaning depends on operating objective (sensitivity/precision), rather than being simply high or low.

Figure 3. Radar Chart (Overall Performance)



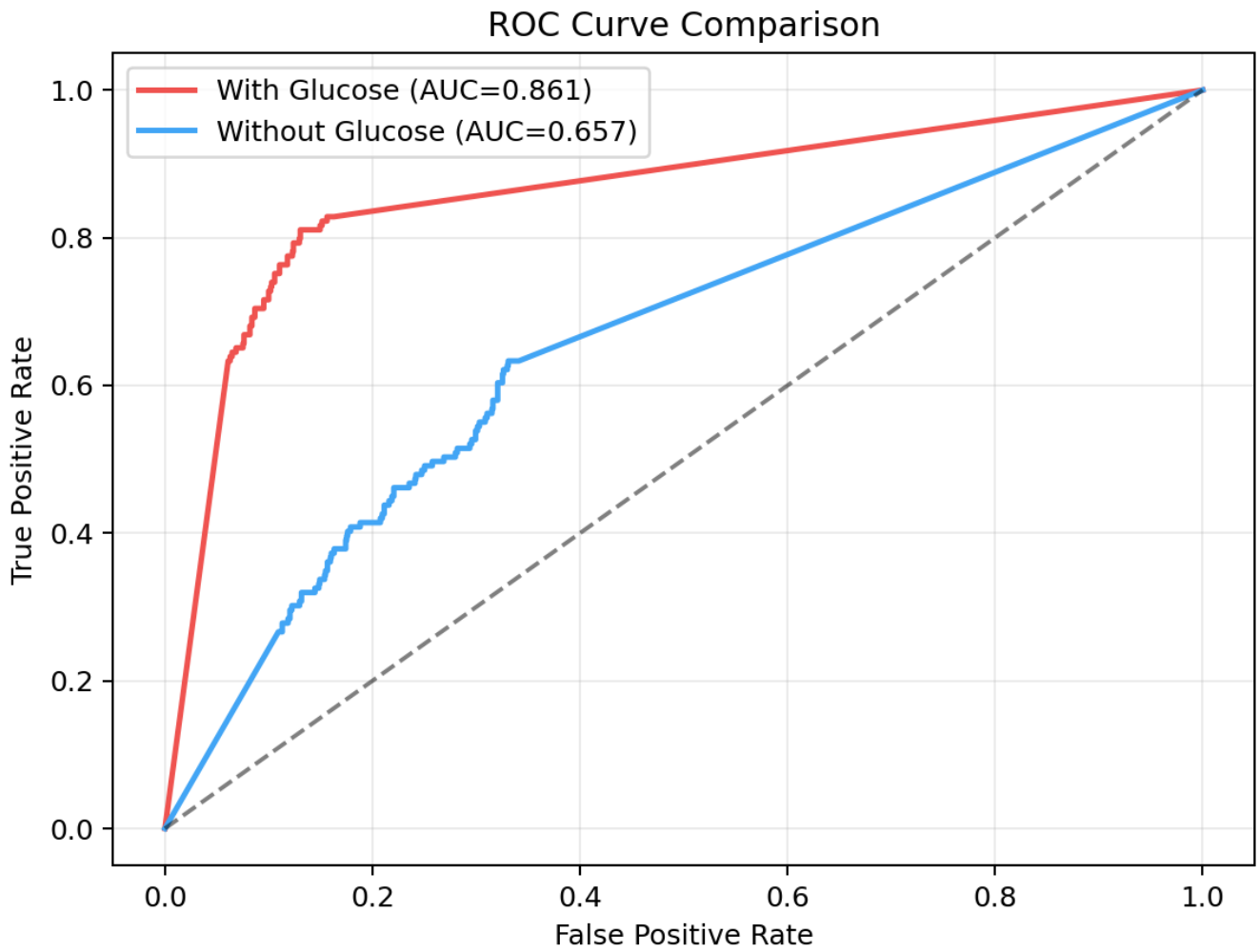
- This chart provides an overall comparison by showing multiple metrics simultaneously in a radial plot.
- Each axis (radial axis): Accuracy, Balanced Accuracy, Precision, Recall, F1, ROC-AUC
- Scores increase from center to outer edge (0→1)
- Interpretation point: wider and more outward area indicates better overall performance. The with-glucose model covers a larger area.

Figure 4. Confusion Matrix Comparison



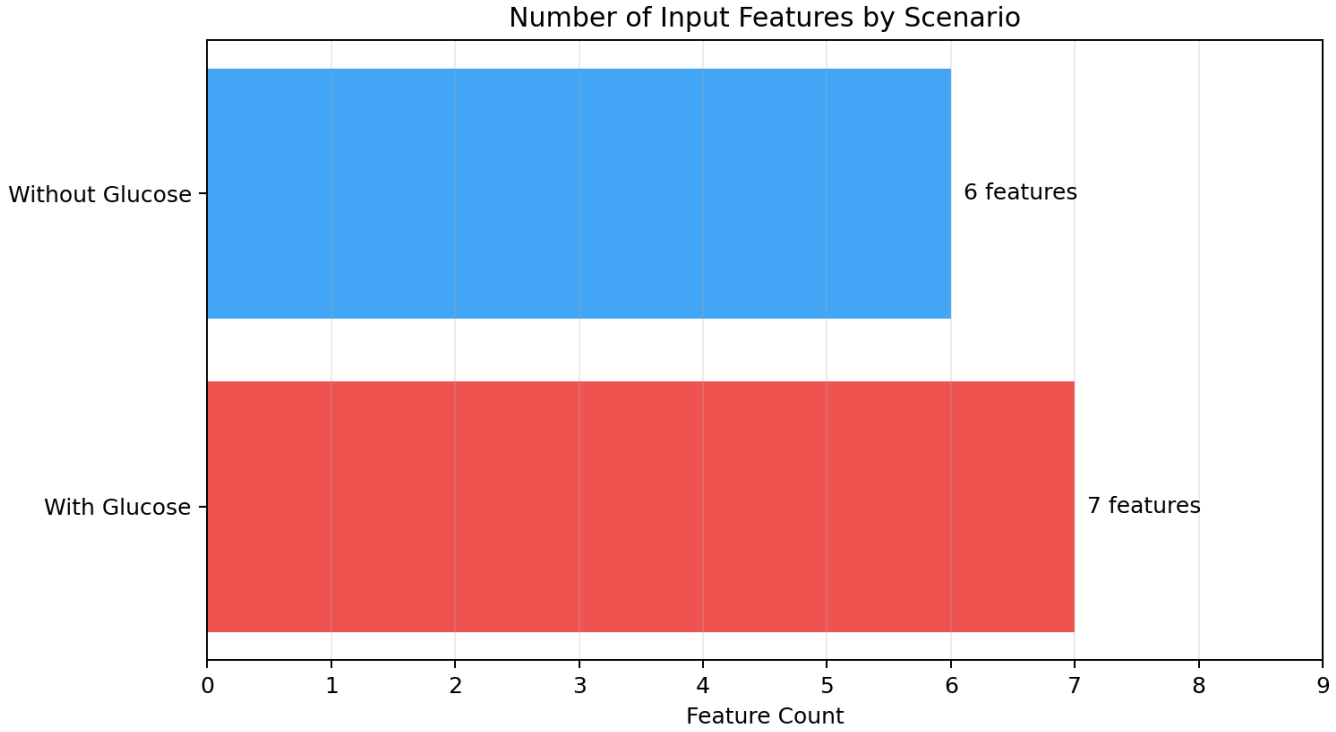
- This chart shows correct/incorrect predictions in a 2x2 table.
- X-axis: model predictions (Pred 0, Pred 1), Y-axis: true labels (True 0, True 1)
- Cell meaning: TN (correct normal), FP (normal predicted as diabetes), FN (diabetes missed as normal), TP (correct diabetes)
- Interpretation point: from a medical perspective, **lower FN is better**. The with-glucose model has lower FN.

Figure 5. ROC Curve Comparison



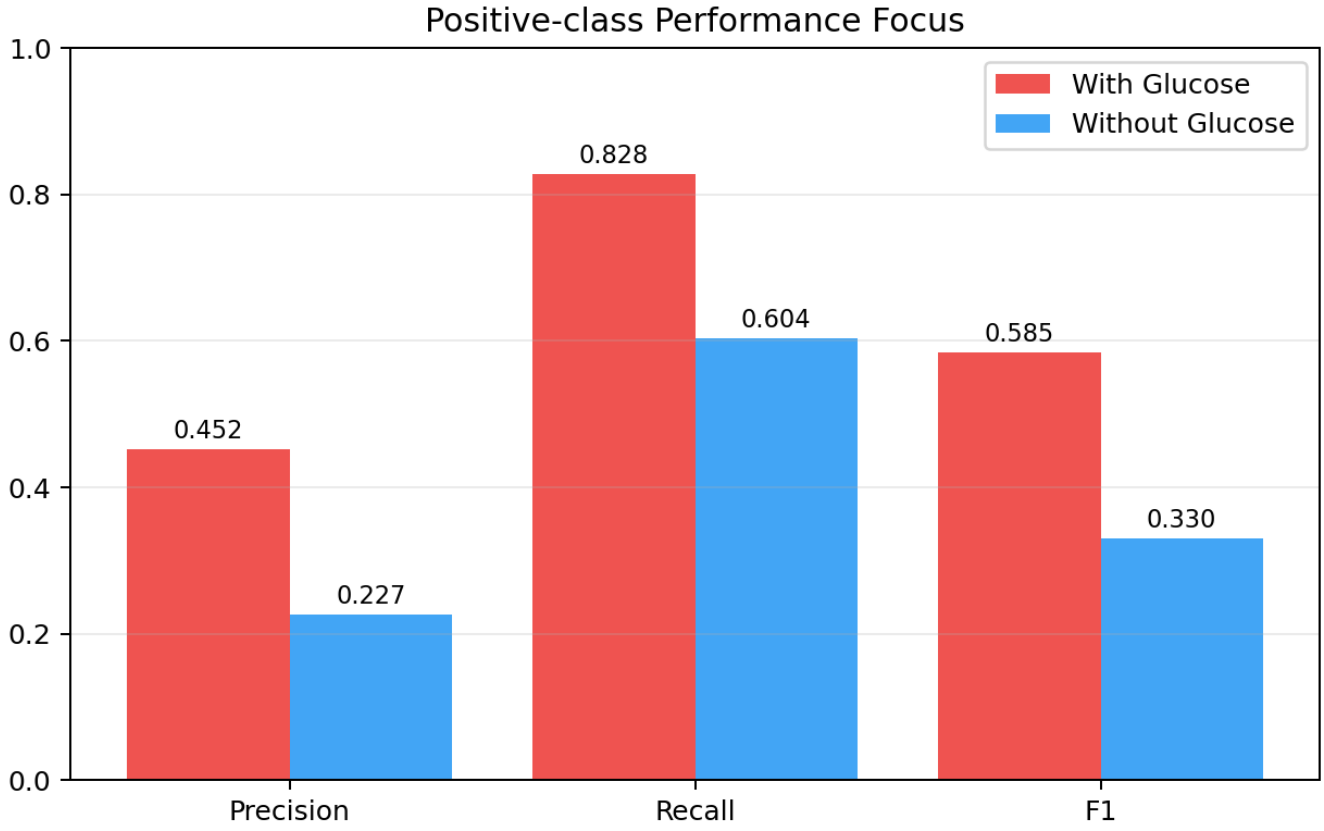
- This chart shows how discriminative performance changes as threshold changes.
- X-axis: FPR (False Positive Rate)
- Y-axis: TPR (True Positive Rate = Recall)
- Interpretation point: curves closer to the **top-left** are better, and higher AUC is better.

Figure 6. Number of Input Features Comparison



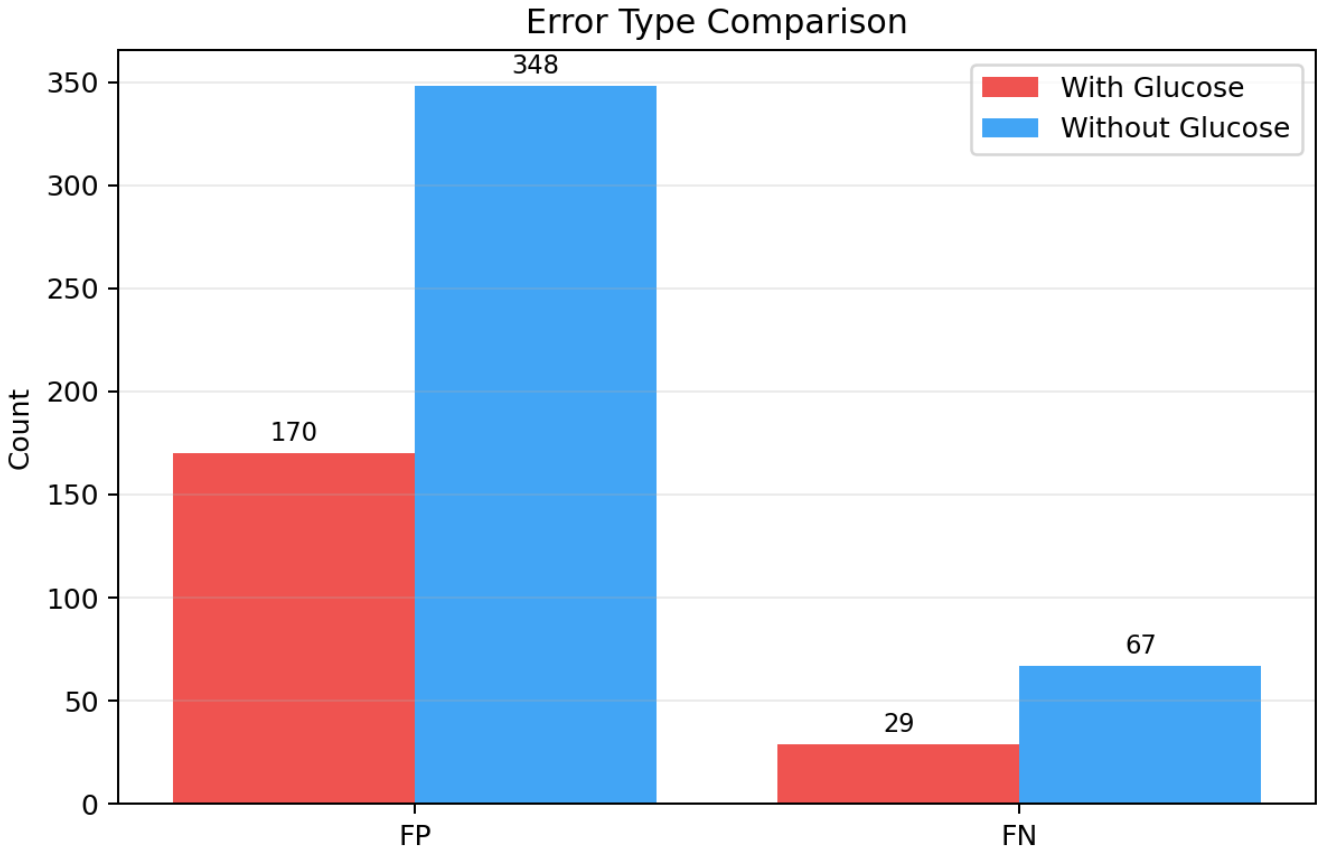
- This chart compares input feature counts by scenario.
- X-axis: number of features
- Y-axis: scenario (with/without glucose)
- Interpretation point: the with-glucose scenario has one additional feature (`HE_glu`).

Figure 7. Precision/Recall/F1 Comparison



- This chart compares only positive-class (diabetes) metrics.
- X-axis: Precision, Recall, F1
- Y-axis: score (0-1)
- Interpretation point: **higher bars are better**. The Recall difference especially highlights patient-detection capability.

Figure 8. Error Type (FP/FN) Comparison



- This chart compares counts of two key model errors (FP, FN).
- X-axis: error type (FP, FN)
- Y-axis: number of errors (count)
- Interpretation point: in medical screening, reducing FN (missed patients) is important. **Lower bars are preferred**.

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12. Reproducibility (Run Again)

```
cd fastapi
source .venv/bin/activate

# Train model with glucose
python train_knhanes.py --with-glucose --feature-eng --poly --smote --score-by
balanced_recall --save

# Train model without glucose
python train_knhanes.py --feature-eng --poly --smote --score-by balanced_recall --save
```

Data file location:

- `fastapi/resources/data/HN19_ALL.sav`

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13. Production Scenario Simulation (Current App Logic)

13-1. Terminology

Term	Definition
`glu_exact`	`HE_glu(공복 혈당)` entered as a continuous value (exact value)

Term	Definition
`glu_binned`	Glucose entered by range (midpoint)
`no_glu`	`HE_glu(공복 혈당)` not entered
`blend`	Probability blending to reduce glucose dependency: `0.55 * no_glu + 0.45 * glu`
`blend_exact`	Glucose + other risk-factor blend (exact glucose input)
`blend_binned`	Glucose + other risk-factor blend (binned glucose input)

13-2. Simulation Result Table (`resources/simulation/simulation_summary.csv`)

> Note: In this table, **prediction rate** means `Accuracy`.
> Note: `FN` is count of missed diabetes cases, and `FP` is count of non-diabetes predicted as diabetes.

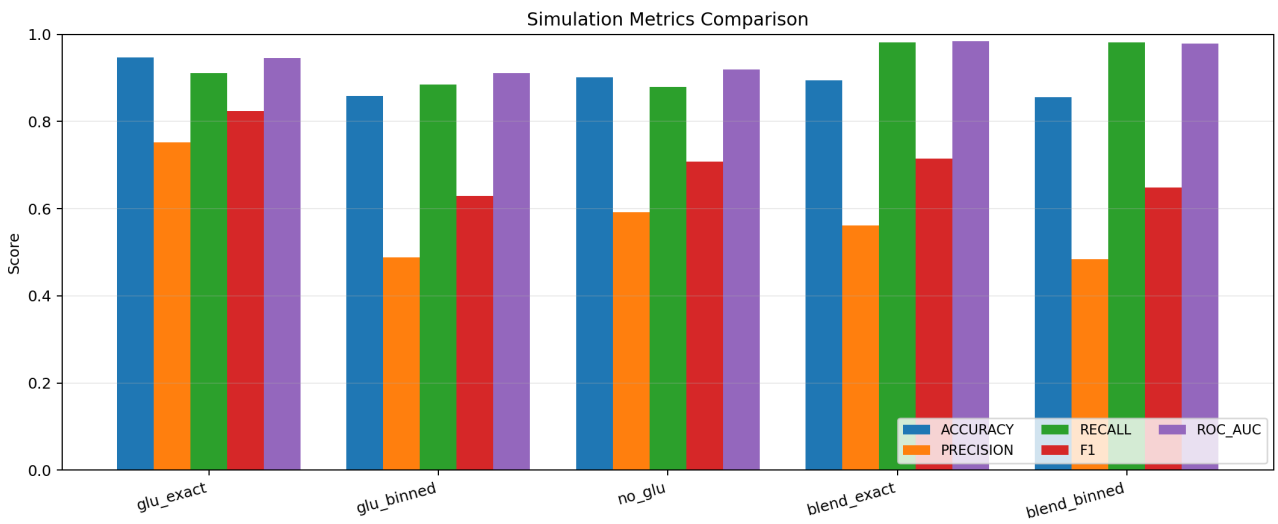
Scenario	Accuracy	Precision	Recall	F1	ROC-AUC	FP	FN
glu_exact	0.9475	0.7526	0.9108	0.8242	0.9444	47	14
glu_binned	0.8590	0.4877	0.8854	0.6290	0.9116	146	18
no_glu	0.9020	0.5923	0.8790	0.7077	0.9156	95	19
blend_exact	0.8942	0.5620	0.9809	0.7146	0.9840	120	3
blend_binned	0.8564	0.4843	0.9809	0.6484	0.9783	164	3

> One-line takeaway: **Highest Accuracy** is `glu_exact (94.75%)`,
> **Highest Recall** is `blend_exact/blend_binned (98.09%)`.

13-3. Interpretation Summary

Perspective	Interpretation
Best overall performance	`glu_exact` is best in Accuracy and F1
Glucose range input	`glu_binned` has information loss compared to exact input
FN-minimizing operation	`blend` is favorable for reducing FN, with an FP increase tradeoff

13-4. Charts



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13-5. Blend Operating Point (Threshold) Tuning Results

> The values below were obtained by tuning only the final classification threshold on blend probabilities, without retraining.
> Applied setting: fixed `GLUCOSE_BLEND_WEIGHT=0.55`, increased `BLEND_THRESHOLD`.

Category	weight	threshold	exact (FP/FN)	binned (FP/FN)
Before tuning (detection-maximizing operating point)	0.55	0.15	120 / 3	164 / 3
After tuning (FP-suppressing operating point)	0.55	0.54	32 / 21	45 / 20

- Interpretation:
  - Increasing threshold greatly reduces false positives (FP).
  - FN (misses) increases in return, so operating point selection should follow objective (FP suppression vs. miss minimization).
- Output files:
  - `resources/simulation/blend_threshold_sweep.csv`
  - `resources/simulation/blend_threshold_tuning.md`

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14. F1-F2 Pre-Validation Results (Additional Feature Plan)

14-1. Planned Feature Mapping Table

Planned Feature	Data Variable (Korean)	Application Status
F1 Family history	`HE_DMfh1(부 당뇨 진단력)`, `HE_DMfh2(모 당뇨 진단력)`, `HE_DMfh3(형제자매 당뇨 진단력)`	Simulation mapping completed
F2 Hypertension/medication	`DI1_dg(고혈압 의사진단)`, `HE_HPdr(검진당일 혈압약 복용)`, `HE_HP(고혈압 유병)`	Simulation mapping completed

14-2. Result Table Including F1/F2 Ablation (`resources/simulation/feature_plan_simulation_summary.csv`)

> Terminology: in `optional mode`, `none` = no optional inputs, `f1` = family history only, `f2` = hypertension/medication only, `f12` = both

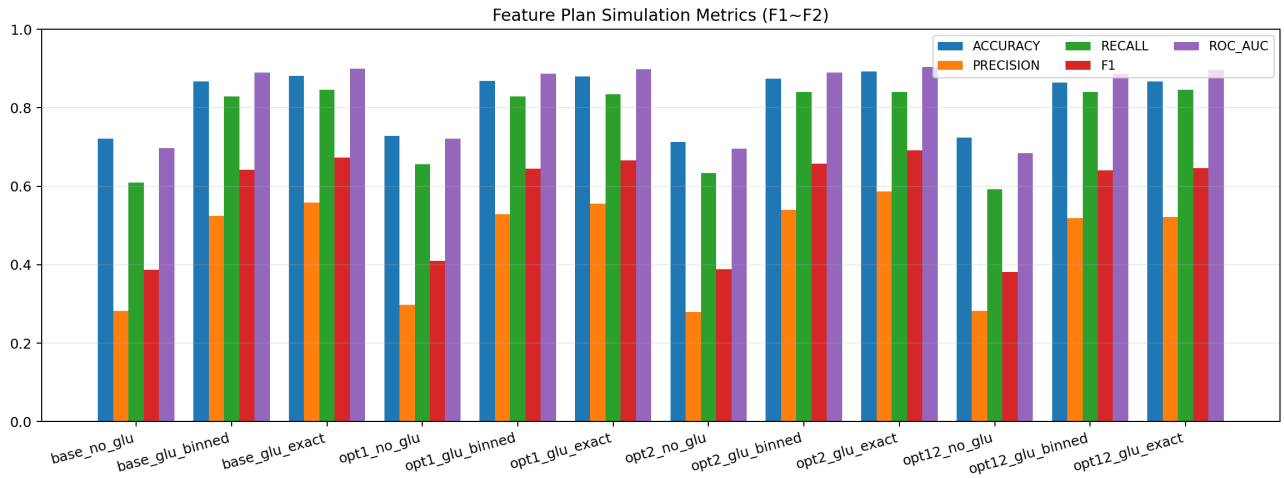
> Terminology: in `glucose mode`, `none` = no glucose, `binned` = binned glucose, `exact` = exact glucose

Scenario	Optional mode	Glucose mode	Accuracy	Precision	Recall	F1	ROC-AUC	FP	FN
base_no_glu	none	none	0.7219	0.2830	0.6095	0.3865	0.6974	261	66
opt1_no_glu	f1	none	0.7287	0.2984	0.6568	0.4104	0.7211	261	58
opt2_no_glu	f2	none	0.7134	0.2801	0.6331	0.3884	0.6955	275	62
opt12_no_glu	f12	none	0.7245	0.2817	0.5917	0.3817	0.6852	255	69
base_glu_binned	none	binned	0.8673	0.5243	0.8284	0.6422	0.8892	127	29
opt1_glu_binned	f1	binned	0.8690	0.5283	0.8284	0.6452	0.8875	125	29
opt2_glu_binned	f2	binned	0.8741	0.5399	0.8402	0.6574	0.8901	121	27
opt12_glu_binned	f12	binned	0.8648	0.5182	0.8402	0.6411	0.8873	132	27
base_glu_exact	none	exact	0.8818	0.5586	0.8462	0.6729	0.8995	113	26
opt1_glu_exact	f1	exact	0.8801	0.5551	0.8343	0.6667	0.8983	113	28
opt2_glu_exact	f2	exact	0.8920	0.5868	0.8402	0.6910	0.9042	100	27
opt12_glu_exact	f12	exact	0.8665	0.5219	0.8462	0.6456	0.8965	131	26

14-3. Interpretation Summary

Segment	Interpretation
No glucose (`none`)	`F1` alone improves Recall (+0.0473) and FN (-8). `F2` slightly improves Recall but increases FP. Simultaneous `F1+F2` is not recommended due to worse Recall/FN
Binned glucose (`binned`)	`F2` alone improves Accuracy/Precision/Recall/F1 (dominant). `F1+F2` improves Recall only, while Accuracy/Precision decrease
Exact glucose (`exact`)	`F2` alone improves Accuracy/Precision/F1, while Recall slightly drops (-0.0060). Benefit of `F1` alone or `F1+F2` is limited

14-4. Charts



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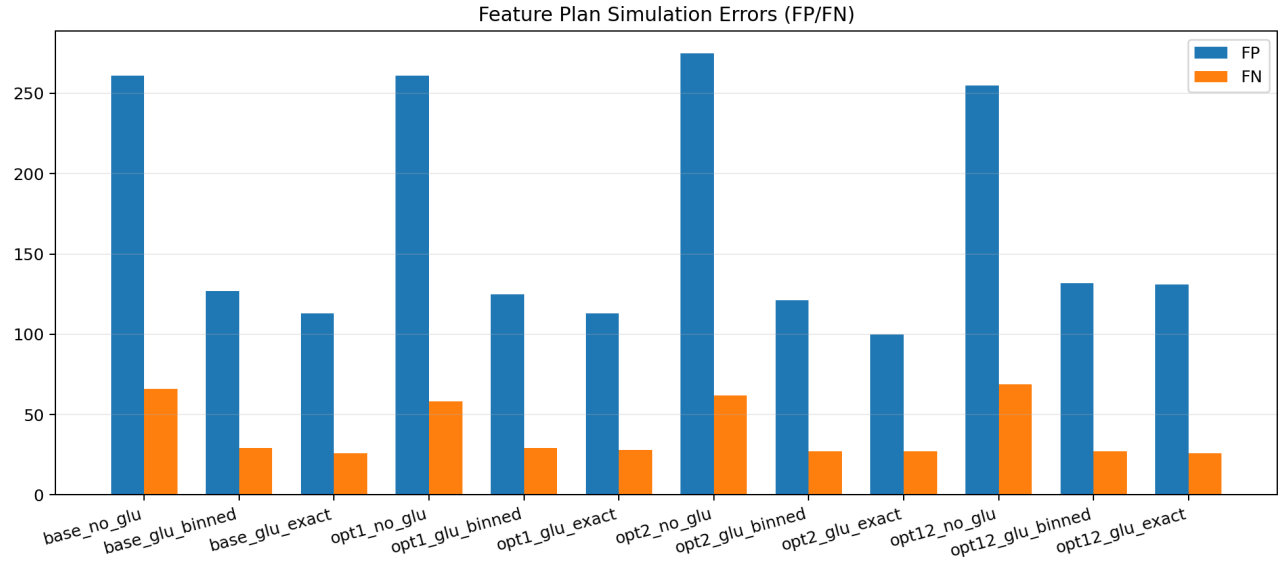

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15. Application Review Outcome and Future Plan

Item	Review Criteria	Summary in This Report
Simple tab	Minimize input burden	Keep current design (no change)
Detailed tab	Performance gain + user acceptability	Conditional extension under review (centered on F1, F2)
F1 family history	Recall/FN improvement confirmed in no_glu segment, limited benefit when glucose is entered	Conditional candidate (priority in no-glucose path)
F2 hypertension/medication	Reproduced Accuracy/F1 gains in binned/exact segments	Priority candidate for review (1st priority in detailed tab)
Waist circumference input	User input feasibility and model stability	Keep as required input
Operational monitoring	Track Recall/FN/FP	Quarterly performance checks

---

16. Additional Reproduction Commands (Simulation)

```
cd fastapi
source .venv/bin/activate

# Production-scenario simulation (including blend)
python simulate_optional_input_cases.py

# Simulation with/without planned features (F1-F2)
python simulate_feature_plan_cases.py

# Operating-point tuning without retraining (blend weight / threshold)
python tune_blend_operating_point.py
```

---

17. Summary and Future Deployment Direction

Item	Summary
Product structure	Keep the simple tab, and gradually review extended items in the detailed tab
Input policy	Keep `HE_wc(허리둘레)` as a required input
Extension priority	F2 (hypertension/medication) first, then conditional review of F1 (family history)
Operations direction	Keep the baseline path; reflect extended paths based on validation outcomes
Operational parameters	Apply `GLUCOSE_BLEND_WEIGHT=0.55`, `BLEND_THRESHOLD=0.54` (for FP suppression)

17-1. Stepwise Rollout Proposal (Draft)

1. **Phase 1**: Review F2-only integration in detailed-tab extended path (especially binned/exact), track Recall/FP/FN
2. **Phase 2**: Review conditional F1 integration focused on no-glucose path and re-evaluate

17-2. Performance Review Criteria

Category	Criteria (Review Metrics)
Improvement judgment	After detailed-tab extension, confirm improved `Recall` or reduced `FN`
Warning signal	Repeated performance drop in exact-glucose segment or sudden FP increase

18. API Dummy Test Cases (Live Responses)

To verify server policies (F2 priority, F1 conditional, automatic exclusion outside KNHANES), we made real `/predict` calls.

- Run date/time: 2026-02-21
- Runtime environment: `uvicorn app.main:app --host 127.0.0.1 --port 8000`
- Common: `chart_image_base64` in responses is omitted in the table

18-1. Case Summary Table

Case	Purpose	HTTP	Prediction result (prediction/probability)	Used model (`used_model`)	Policy verification point
case1_knhanes_no_glu_f1f2	No glucose + F1/F2 both entered	200	1 / 1.0	KNHANES without glucose	`input` includes both `family_history_dm` and `htn_or_med`
case2_knhanes_glu_f1f2	Glucose entered + F1/F2 both entered	200	1 / 0.55	KNHANES blend	`family_history_dm` excluded from `input`, `htn_or_med` retained
case3_knhanes_glu_f2_only	Glucose entered + F2 only	200	1 / 0.1607	KNHANES blend	`input` includes only `htn_or_med`

Case	Purpose	HTTP	Prediction result (prediction/probability)	Used model (`used_model`)	Policy verification point
case4_pima_with_f1f2	Pima path + F1/F2 entered	200	0 / 0.4062	AdaBoost (with glucose)	F1/F2 automatically excluded from `input`
case5_range_error	Range error (glucose 300)	400	-	-	Verify range-validation error message
case6_minimal_no_glu	Minimal input (no glucose)	200	1 / 1.0	KNHANES without glucose	Verify normal response in baseline path

18-2. Key Live Response Excerpts (JSON)

case2_knhanes_glu_f1f2 (Glucose entered + F1/F2 both entered)

```
{
  "prediction": 1,
  "probability": 0.55,
  "used_model": "KNHANES blend (55% risk factors + 45% glucose)",
  "input": {
    "glucose": 95.0,
    "bmi": 28.0,
    "age": 47.0,
    "waist_cm": 94.0,
    "sex": 1.0,
    "height_cm": 170.0,
    "htn_or_med": 0.0
  }
}
```

Verification: Although `family history` was included in the request, it was excluded from response `input`, confirming server policy application.

case4_pima_with_f1f2 (Pima path + F1/F2 entered)

```
{
  "prediction": 0,
  "probability": 0.4062,
  "used_model": "AdaBoost (with glucose)",
  "input": {
    "pregnancies": 2.0,
    "glucose": 95.0,
    "bmi": 28.0,
    "age": 47.0
  }
}
```

Verification: In the Pima path, `family_history_dm` and `htn_or_med` are automatically excluded.

case5_range_error (Glucose range error)

```
{
  "detail": "glucose must be within the range 44.0 to 199.0."
}
```

Verification: Input range validation works as expected.